```
import pandas as pd

# File name
CSV_PATH = "placement_predict_50k_Dataset.csv"

# Read CSV file
df = pd.read_csv('/content/placement_predict_50k Dataset.csv')

# 1. Display first 5 rows
print("FIRST 5 ROWS")
print(df.head())

# 2. Display dataset shape print("\nDATASET SHAPE") print(df.shape)

# 3. Display column names print("\nCOLUMN NAMES") print(df.columns)

# 4. Display data types print("\nDATA TYPES") print(df.dtypes)

# 5. Display basic information print("\nDATASET INFORMATION")
df.info()

# 6. Display statistical summary print("\nSTATISTICAL SUMMARY")
print(df.describe())

# 7. Check missing values print("\nMISSING VALUES")
print(df.isnull().sum())

# 8. Check duplicate rows print("\nDUPLICATE ROWS")
print(df.duplicated().sum())

# 9. Placement status distribution print("\nPLACEMENT STATUS DISTRIBUTION")
print(df["PlacementStatus"].value_counts())

# 10. Placement status percentage print("\nPLACEMENT STATUS PERCENTAGE")
print(df["PlacementStatus"].value_counts(normalize=True) * 100)
```

```
FIRST 5 ROWS
   StudentID  Gender       City CollegeTier Stream Specialisation Hostel  \
0          1    Male  Ahmedabad       Tier2    ECE     Networking     No
1          2  Female     Mumbai       Tier2    ECE    DataScience    Yes
2          3    Male    Kolkata       Tier2     IT    DataScience    Yes
3          4    Male     Jaipur       Tier1     CS             AI     No
4          5    Male       Pune       Tier2     IT    DataScience    Yes

  HistoryOfBacklogs  SGPA_Sem1  SGPA_Sem2  ...  Publications  \
0                No       6.02       6.54  ...             0
1               Yes       5.84       5.12  ...             0
2                No       4.91       5.29  ...             0
3                No       7.67       8.03  ...             0
4                No       8.14       8.97  ...             1

   AptitudeTestScore  SoftSkillsRating  CodingTestScore  MockInterviewScore  \
0               66.7               2.2             49.4                47.8
1               48.2               2.4             26.7                25.8
2               73.8               2.8             67.7                41.5
3               69.8               2.7             66.9                48.0
4               73.1               2.1             71.7                61.7

   ExtraCurricular  CGPA_Tier  PlacementStatus  IsAnomaly  Salary Package
0                0        Low                0          0            0.00
1                0        Low                0          0            0.00
2                0        Low                1          0            3.89
3                0        Mid                1          0            8.37
4                1       High                1          0           18.99

[5 rows x 32 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 32 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   StudentID           50000 non-null  int64
 1   Gender              50000 non-null  object
 2   City                50000 non-null  object
 3   CollegeTier         50000 non-null  object
 4   Stream              50000 non-null  object
 5   Specialisation      50000 non-null  object
 6   Hostel              50000 non-null  object
 7   HistoryOfBacklogs   50000 non-null  object
 8   SGPA_Sem1           50000 non-null  float64
 9   SGPA_Sem2           50000 non-null  float64
 10  SGPA_Sem3           50000 non-null  float64
 11  SGPA_Sem4           50000 non-null  float64
```

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 12  SGPA_Sem5           50000 non-null  float64
 13  SGPA_Sem6           50000 non-null  float64
 14  SGPA_Sem7           50000 non-null  float64
 15  SGPA_Sem8           50000 non-null  float64
 16  CGPA                50000 non-null  float64
 17  AttendancePercent   50000 non-null  float64
 18  Internships         50000 non-null  int64
 19  Projects            50000 non-null  int64
 20  Workshops           45512 non-null  float64
 21  Certifications      50000 non-null  int64
```